RAG Learning

Topic 5 — Separate Chemistry 1

Edexcel GCSE (9-1) Chemistry (1CH0) — Paper 1

Topic 5 gathers together the extra content that only appears in the separate science GCSE. It has four strands: **transition metals, alloys and corrosion**; **quantitative analysis** (concentrations, titrations, percentage yield, atom economy and molar gas volume); **dynamic equilibria** applied to industry (the Haber process conditions and fertilisers); and **chemical cells and fuel cells**. It builds directly on Topic 1 (moles and formula masses) and Topic 4 (reactivity, the Haber process and equilibrium), so revise those first.

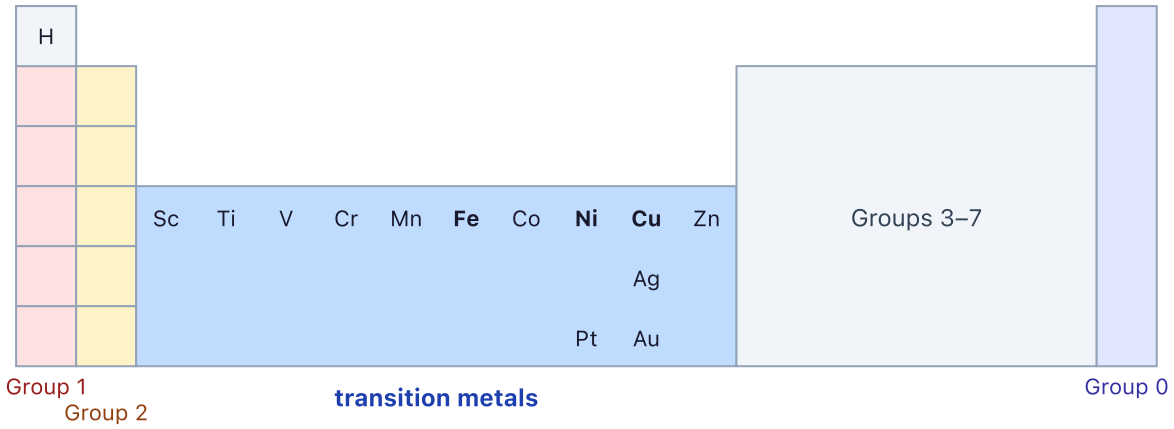
The whole of Topic 5 is chemistry only — every point carries a "C" suffix in the specification (5.1C–5.27C) and is *not* examined in Combined Science. Content marked **(HT)** is Higher Tier only. The topic contains **one core practical**: an accurate acid–alkali titration (CP5, spec 5.9C).

5.1 Transition Metals

> The **transition metals** are the block of metals in the centre of the periodic table, between Group 2 and Group 3. **Most metals are transition metals**. Examples include iron (Fe), copper (Cu), nickel (Ni), chromium (Cr), manganese (Mn), cobalt (Co), zinc (Zn), silver (Ag), gold (Au) and platinum (Pt).

© RAG Learning · Not for redistribution

Where the transition metals are



Typical properties of transition metals

Transition metals have the typical properties of metals (shiny, good conductors of heat and electricity, malleable, ductile) and in addition:

- **High melting points** (except mercury) — e.g. iron melts at about 1540 °C.
- **High densities** — e.g. iron about 7.9 g cm^{-3}, copper about 8.9 g cm^{-3}.
- They form **coloured compounds**.
- The metals and their compounds show **catalytic activity** — e.g. **iron in the Haber process**.

Property	Transition metals (e.g. Fe, Cu)	Group 1 metals (e.g. Na, K)
Melting point	High (Fe ~1540 °C, Cu ~1085 °C)	Low (Na 98 °C, K 63 °C)
Density	High (Fe 7.9 g cm^{-3})	Low (Na 0.97 g cm^{-3} — floats on water)
Hardness / strength	Hard and strong	Soft — can be cut with a knife
Reactivity with water	Slow or none	Rapid, forms hydroxide + hydrogen
Colour of compounds	Often coloured	White solids, colourless solutions
Catalytic activity	Common	Not used as catalysts

© RAG Learning · Not for redistribution

Coloured compounds and catalysts

Ion / compound	Colour in solution
Copper(II), Cu^{2+} (e.g. $CuSO_4$)	Blue
Iron(II), Fe^{2+}	Pale green
Iron(III), Fe^{3+}	Yellow / orange-brown
Nickel(II), Ni^{2+}	Green
Manganate(VII), MnO_4^-	Purple

Catalyst	Process
Iron	Haber process — making ammonia (the example named in the specification)
Nickel	Hydrogenation of vegetable oils to make margarine
Vanadium(V) oxide	Contact process — making sulfuric acid
Manganese(IV) oxide	Decomposition of hydrogen peroxide
Platinum, rhodium	Catalytic converters in car exhausts

Exam Tip: Coloured compounds are a good way to **identify** a transition metal in a "which element is this?" question: a high-melting, dense metal that forms a blue sulfate and catalyses a reaction is a transition metal, not a Group 1 or Group 2 metal.

5.2 Corrosion and Rusting

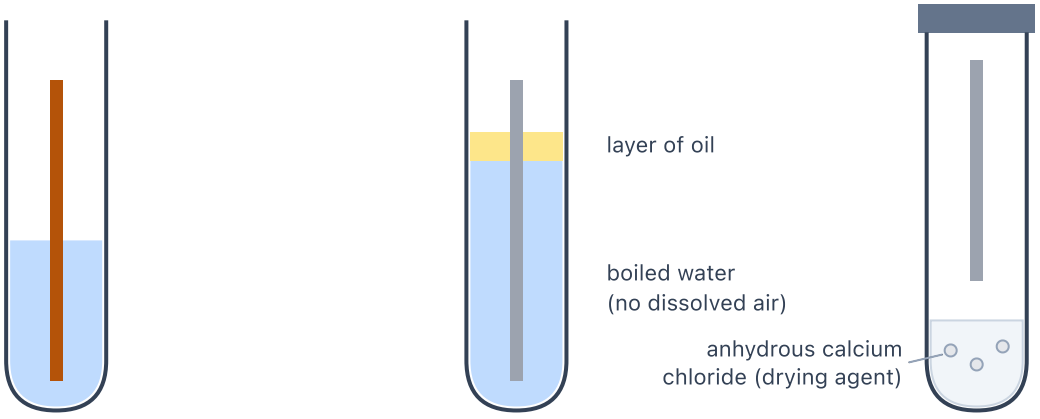
© RAG Learning · Not for redistribution

Corrosion is the gradual destruction of a metal caused by its **oxidation**, usually by reaction with oxygen (and often water) in the environment. **Rusting** is the corrosion of **iron** (and steel): it needs **both oxygen and water**.

iron + oxygen + water → hydrated iron(III) oxide (rust)

In rusting, iron atoms lose electrons to form iron ions — the iron is **oxidised** (it gains oxygen and loses electrons). Rust is flaky and crumbles away, exposing fresh iron underneath, so rusting continues until the iron is destroyed. Salt water speeds up rusting.

What does iron need to rust?



A
air + water
RUSTS

B
water, no air
no rust

C
dry air, no water
no rust

Preventing rusting

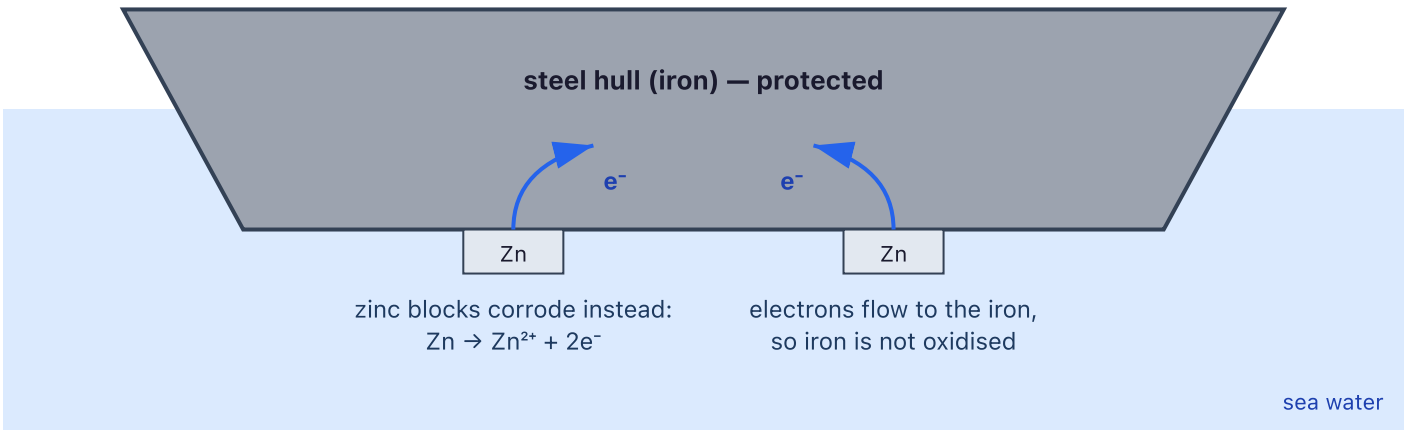
The specification asks you to explain three approaches: excluding oxygen, excluding water, and sacrificial protection.

Method	How it works	Examples
Barrier methods — painting, oiling/greasing, coating with plastic	Form a layer that **excludes oxygen and water** from the iron surface	Paint on cars and bridges; grease on bike chains and machinery; plastic-coated garden wire

© RAG Learning · Not for redistribution

Electroplating / tin-plating	A thin layer of a less reactive metal acts as a barrier	Tin-plated steel food cans; chromium-plated taps
Storing with a **drying agent**	Removes water vapour from the air	Silica gel packets with tools
Sacrificial protection	A **more reactive** metal (zinc or magnesium) is attached to or coated on the iron. It is **oxidised in preference** to iron, so the iron does not rust even if exposed.	Magnesium or zinc blocks bolted to ships' hulls and underground pipes
Galvanising	Coating iron with **zinc**. Acts as a barrier *and* provides sacrificial protection if scratched.	Galvanised buckets, fences, car bodies, lamp posts

Sacrificial protection of a ship's hull



Worked Example: Explain why a scratched galvanised (zinc-coated) bucket does not rust, but a scratched tin-plated can does.

Galvanised: zinc is **more reactive** than iron, so where the iron is exposed the zinc is oxidised instead ($Zn \rightarrow Zn^{2+} + 2e^-$); the zinc is sacrificed and the iron stays protected.
Tin-plated: tin is **less reactive** than iron, so it only works as a barrier. Once scratched, oxygen and water reach the iron and it rusts — in fact faster than unprotected iron, because the iron now loses electrons in preference to the tin.

© RAG Learning · Not for redistribution

Exam Tip: For sacrificial protection you must say the protecting metal is **more reactive** than iron and is **oxidised instead of** iron (or "loses electrons more readily"). "The zinc protects the iron" alone scores nothing. Remember the sacrificial metal must be **replaced** when it has corroded away.

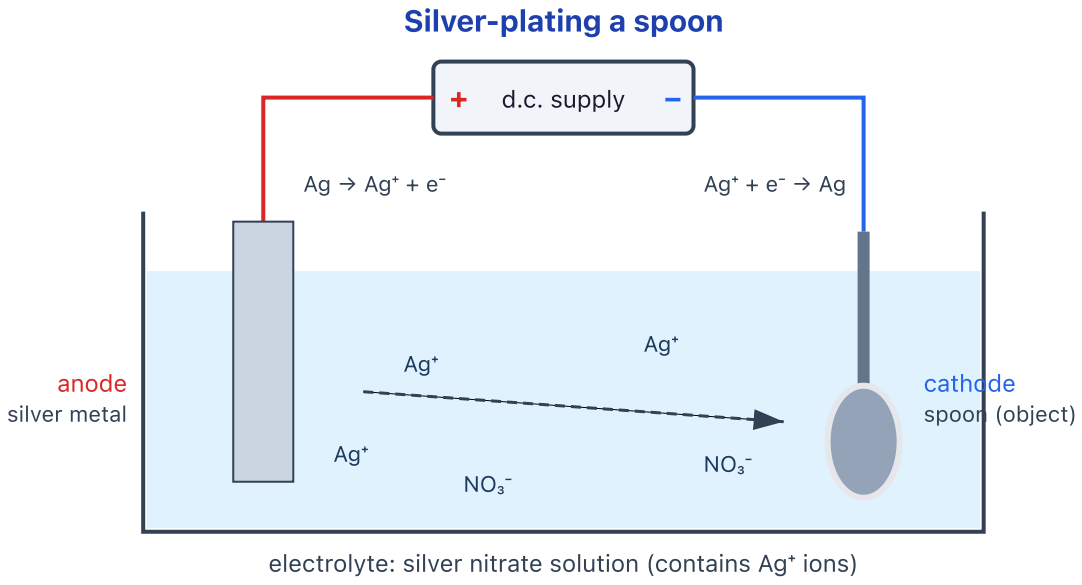
5.3 Electroplating

Electroplating uses electrolysis to coat a metal object with a thin layer of another metal. It is used to improve **appearance** (e.g. silver or gold plating) and/or **resistance to corrosion** (e.g. chromium or tin plating).

The set-up:

- The **object to be plated** is the **cathode** (negative electrode).
- The **anode** (positive electrode) is made of the **plating metal**.
- The **electrolyte** is a solution containing **ions of the plating metal**.

At the cathode, metal ions gain electrons and are deposited on the object. At the anode, metal atoms lose electrons and go into solution, replacing the ions used up — so the concentration of the electrolyte stays the same.



Plating metal	Purpose	Example
Silver / gold	Attractive appearance at low cost; resists tarnishing	Cutlery, jewellery
Chromium	Shiny and hard; resists corrosion	Taps, bicycle handlebars
Tin	Unreactive, non-toxic barrier to corrosion	Steel food cans
Nickel	Corrosion resistance; base layer for chromium	Coins, fittings

Exam Tip: If you are asked to "describe how to copper-plate a key", name all three parts: **key = cathode**, **copper = anode**, **copper(II) sulfate solution = electrolyte**, connected to a d.c. supply. **(HT)** Then add the cathode half equation $Cu^{2+} + 2e^- \rightarrow Cu$.

5.4 Alloys

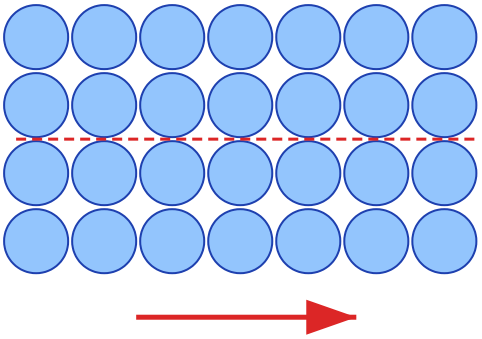
An **alloy** is a mixture of a metal with one or more other elements (usually other metals, sometimes carbon). Converting a pure metal into an alloy often **increases its strength** (and hardness).

Why alloys are stronger — the model

In a **pure metal** all the atoms are the same size and are arranged in regular **layers**. When a force is applied, the layers can **slide over each other** easily, so pure metals are soft and malleable. In an **alloy**, atoms of different sizes **distort the regular layers**, so it is much more difficult for the layers to slide — the alloy is **stronger and harder**.

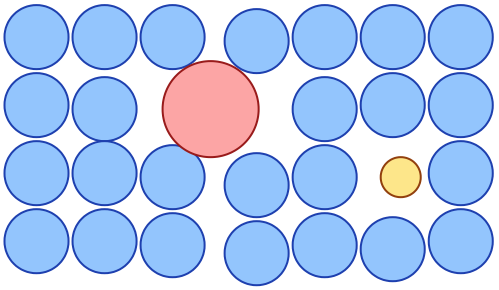
© RAG Learning · Not for redistribution

Pure metal



regular layers slide easily → soft, malleable

Alloy



different-sized atoms distort the layers → layers cannot slide easily → stronger, harder

Exam Tip: The marking points are: pure metal → **regular layers** of identical atoms → layers **slide**; alloy → **different-sized atoms** → **disrupt / distort the layers** → harder for layers to slide → stronger. Do not say the bonds in an alloy are "stronger".

Alloy steels

Pure iron is too **soft** for most uses, and the iron straight from the blast furnace (cast iron, about 4% carbon) is **brittle**. Iron is therefore made into **steels** by controlling the carbon content and alloying it with other metals such as chromium, nickel, manganese, tungsten or vanadium. Each alloy steel has properties suited to a particular use.

Steel	Composition	Properties	Uses
Low-carbon (mild) steel	Iron + up to about 0.3% carbon	Strong, easily shaped (malleable), but rusts	Car bodies, girders, wire, nails
High-carbon steel	Iron + about 0.6–1.5% carbon	Very hard and strong but more brittle	Cutting tools, drill bits, railway lines
Stainless steel	Iron + **chromium** (and often **nickel**)	Hard, strong and **resistant to corrosion**	Cutlery, sinks, surgical instruments, chemical plant

© RAG Learning · Not for redistribution

Tungsten / manganese steels	Iron + tungsten or manganese	Very hard, keep hardness at high temperature / tough	High-speed cutting tools; rock-crushing machinery

Uses of metals and alloys related to their properties

Metal / alloy	Key properties	Uses
Aluminium	Low density; resists corrosion (protective oxide layer); good conductor of heat and electricity; malleable	Overhead power cables (light, with a steel core for strength), drinks cans, cooking foil, window frames; aircraft (as alloys)
Copper	Excellent electrical and thermal conductor; ductile; unreactive (does not react with water)	Electrical wiring, water pipes, saucepan bases
Gold	Very unreactive so never tarnishes; shiny; very malleable; good conductor; but soft	Jewellery, coins, electrical connectors in computers, dental work
Magnalium (aluminium + magnesium)	Lower density, stronger and more corrosion-resistant than pure aluminium	Aircraft and car parts, bicycle frames; high-magnesium versions are brittle and used in fireworks
Brass (copper + zinc)	Harder and stronger than copper; resists corrosion; attractive gold colour; low friction	Musical instruments, door handles, screws, plug pins, plumbing fittings
Gold alloys (gold + copper/silver)	Harder and more hard-wearing than pure gold	Jewellery — 24 carat is pure gold; 18 carat is 18/24 = 75% gold

Worked Example: A ring is made of 9 carat gold and has a mass of 4.0 g. Calculate the mass of pure gold in the ring.

© RAG Learning · Not for redistribution

Fraction of gold = 9 ÷ 24 = 0.375 (37.5%)
Mass of gold = 0.375 × 4.0 g = **1.5 g**

Exam Tip: "Explain why aluminium is used for overhead power cables" needs **property → reason**: it is a good *electrical conductor* (carries current), has a *low density* (light, so pylons can be further apart) and *resists corrosion* (lasts outdoors). A single word list with no reasons will not get full marks.

5.5 Concentration of Solutions

Concentration is the amount of solute dissolved in a given volume of solution. It can be measured in **g dm⁻³** (mass per volume) or **mol dm⁻³** (moles per volume). 1 dm³ = 1000 cm³.

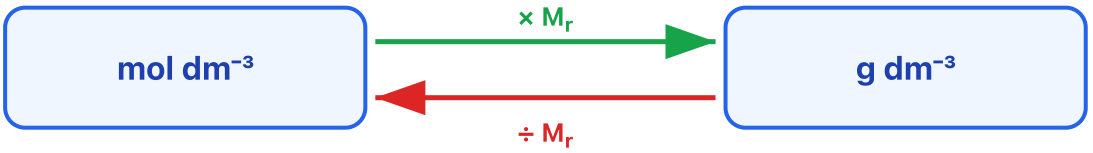
$$\text{concentration (g dm}^{-3}\text{)} = \text{mass of solute (g)} \div \text{volume of solution (dm}^3\text{)}$$

(HT) Calculating concentrations in mol dm⁻³, and converting between g dm⁻³ and mol dm⁻³ (spec 5.8C), is Higher tier only:

$$\text{concentration (mol dm}^{-3}\text{)} = \text{amount of solute (mol)} \div \text{volume of solution (dm}^3\text{)}$$

$$\text{g dm}^{-3} = \text{mol dm}^{-3} \times M_r \quad \leftrightarrow \quad \text{mol dm}^{-3} = \text{g dm}^{-3} \div M_r$$

Converting between concentration units



Always convert cm³ to dm³ first: divide by 1000.

Worked Example: (HT) 5.85 g of sodium chloride (M_r = 58.5) is dissolved in water to make 250 cm^3 of solution. Calculate the concentration in (a) g dm^{-3} and (b) mol dm^{-3}.

Volume = 250 ÷ 1000 = 0.250 dm^3
(a) concentration = 5.85 ÷ 0.250 = **23.4 g dm^{-3}**
(b) moles = 5.85 ÷ 58.5 = 0.100 mol; concentration = 0.100 ÷ 0.250 = **0.400 mol dm^{-3}**
Check: 0.400 × 58.5 = 23.4 g dm^{-3} ✓

Worked Example: (HT) A solution of sulfuric acid, H_2SO_4, has a concentration of 4.9 g dm^{-3}. Convert this to mol dm^{-3}. (A_r: H = 1, S = 32, O = 16)

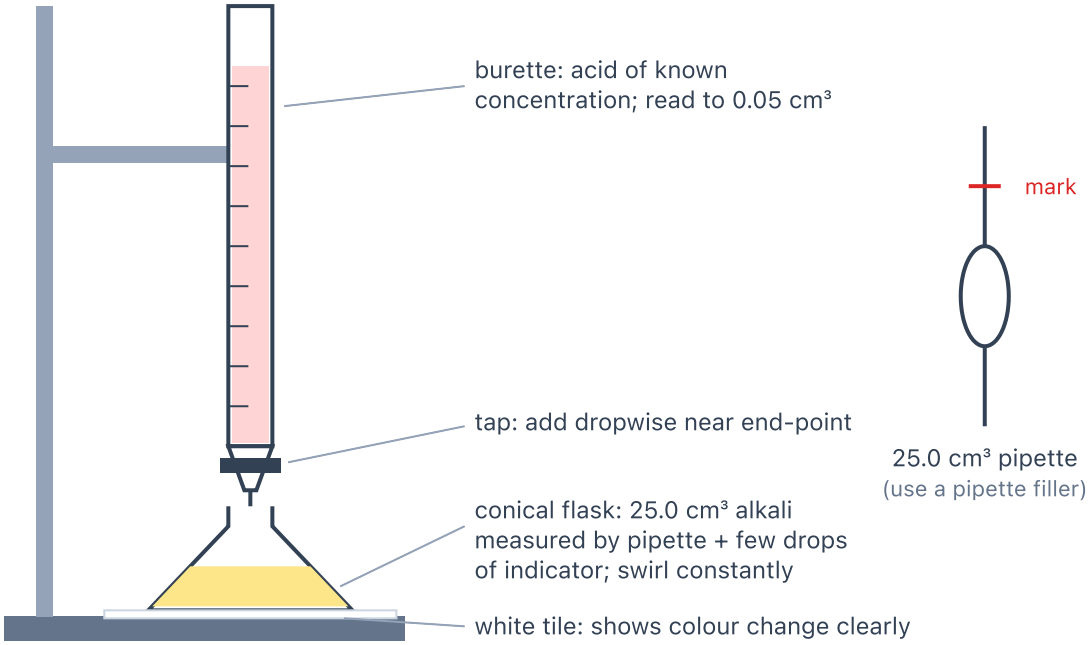
M_r = (2 × 1) + 32 + (4 × 16) = 98
concentration = 4.9 ÷ 98 = **0.050 mol dm^{-3}**

5.6 Core Practical: Acid–Alkali Titration (CP5)

A **titration** is used to find the exact volume of one solution that reacts with a measured volume of another. If the concentration of one is known, the other can be calculated.

© RAG Learning · Not for redistribution

Titration apparatus



Method

1. Rinse the pipette with the alkali and the burette with the acid (removes water/other substances that would dilute them).
2. Use a **pipette and pipette filler** to transfer **25.0 cm³** of alkali (e.g. sodium hydroxide) into a conical flask. Read the bottom of the meniscus on the mark at eye level.
3. Add a few drops of a suitable **indicator** (e.g. **methyl orange** or **phenolphthalein**) and place the flask on a **white tile**.
4. Fill the burette with acid using a funnel (then remove the funnel), make sure the jet below the tap is filled, and record the **initial reading** to the nearest 0.05 cm³.
5. Add acid while **swirling** the flask. Do a quick **rough (range-finder) titration** first.
6. Repeat accurately: near the expected end-point add acid **drop by drop** until the indicator **just** changes colour permanently. Record the **final reading**.
7. Calculate the **titre** (final – initial). Repeat until you have **concordant** titres (within **0.10 cm³** of each other) and calculate the **mean** of the concordant titres only.

Indicator	Colour in alkali	Colour in acid	Use?

© RAG Learning · Not for redistribution

Phenolphthalein	Pink	Colourless	Yes — sharp change
Methyl orange	Yellow	Red (orange at end-point)	Yes — sharp change
Universal indicator / litmus	Range of colours	Range of colours	**No** — gradual colour change gives no sharp end-point

Recording and processing results

	Rough	Run 1	Run 2	Run 3
Final reading / cm^3	23.10	22.45	44.60	22.60
Initial reading / cm^3	0.00	0.00	22.45	0.20
Titre / cm^3	23.10	22.45	22.15	22.40

Run 2 (22.15 cm^3) is not within 0.10 cm^3 of the others, so it is not concordant (probably the end-point was misjudged). Concordant titres: 22.45 and 22.40 cm^3. Mean = (22.45 + 22.40) ÷ 2 = 22.425 = **22.43 cm^3**. The rough titre is **never** included in the mean.

> **Exam Tip:** Precision answers: a **pipette** measures a fixed volume very accurately (better than a measuring cylinder); a **burette** delivers variable volumes accurately; the **white tile** makes the colour change easier to see; **swirling** mixes the reactants; adding **dropwise** avoids overshooting the end-point. To make a *pure salt* from the titration, repeat with the same volumes but **no indicator**, then evaporate and crystallise.

5.7 Titration Calculations (HT)

Use the results of a titration to find an **unknown concentration** or an **unknown volume**. Follow four steps every time:

1. Write the **balanced equation** and note the **mole ratio**.

© RAG Learning · Not for redistribution

2. Calculate the **moles** of the substance whose volume and concentration you know: moles = concentration × volume (dm^3).
3. Use the ratio to find the **moles** of the other substance.
4. Calculate the unknown: concentration = moles ÷ volume (dm^3), or volume = moles ÷ concentration.

Worked Example: (HT) 25.0 cm^3 of sodium hydroxide solution is neutralised by 21.40 cm^3 of 0.100 mol dm^{-3} hydrochloric acid. Calculate the concentration of the sodium hydroxide in mol dm^{-3} and in g dm^{-3}. (M_r NaOH = 40)

$NaOH + HCl \rightarrow NaCl + H_2O$ (ratio 1 : 1)
moles HCl = 0.100 × (21.40 ÷ 1000) = 0.002140 mol
moles NaOH = 0.002140 mol
concentration NaOH = 0.002140 ÷ (25.0 ÷ 1000) = **0.0856 mol dm^{-3}**
in g dm^{-3}: 0.0856 × 40 = **3.42 g dm^{-3}**

Worked Example: (HT) 25.0 cm^3 of 0.150 mol dm^{-3} sodium hydroxide is neutralised by 18.75 cm^3 of dilute sulfuric acid. Calculate the concentration of the acid.

$H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$ (ratio 1 : 2)
moles NaOH = 0.150 × 0.0250 = 0.00375 mol
moles H_2SO_4 = 0.00375 ÷ 2 = 0.001875 mol
concentration H_2SO_4 = 0.001875 ÷ 0.01875 = **0.100 mol dm^{-3}**

Worked Example: (HT) What volume of 0.200 mol dm^{-3} hydrochloric acid is needed to neutralise 20.0 cm^3 of 0.150 mol dm^{-3} potassium hydroxide?

$KOH + HCl \rightarrow KCl + H_2O$ (1 : 1)
moles KOH = 0.150 × 0.0200 = 0.00300 mol = moles HCl
volume HCl = 0.00300 ÷ 0.200 = 0.0150 dm^3 = **15.0 cm^3**

© RAG Learning · Not for redistribution

Exam Tip: The two classic errors are forgetting to divide cm^3 by 1000, and ignoring the mole ratio for **sulfuric acid** (H_2SO_4 reacts with **2** NaOH). Give answers to 3 significant figures, matching the data.

5.8 Percentage Yield

The **theoretical yield** is the maximum mass of product that could be made from the reactants, calculated from the balanced equation. The **actual yield** is the mass of product actually obtained.

percentage yield = (actual yield ÷ theoretical yield) × 100

Why the actual yield is less than the theoretical yield

- **Incomplete reactions** — some reactant does not react (e.g. a reversible reaction reaches equilibrium, or the reaction is stopped too soon).
- **Practical losses** during the experiment — product left behind in apparatus, lost when filtering, transferring liquids or during evaporation and crystallisation.
- **Competing, unwanted reactions (side reactions)** — some reactants form different products.

Worked Example: 50.0 g of calcium carbonate is heated: $CaCO_3 \rightarrow CaO + CO_2$. 22.4 g of calcium oxide is obtained. Calculate the percentage yield. (M_r: $CaCO_3$ = 100, CaO = 56)

From the equation, 100 g of $CaCO_3$ gives 56 g of CaO, so 50.0 g gives a theoretical yield of 56 × (50.0 ÷ 100) = 28.0 g
percentage yield = (22.4 ÷ 28.0) × 100 = **80.0%**

Worked Example: A student expects to make 12.0 g of magnesium sulfate. The percentage yield is 65%. What mass did the student actually obtain?

© RAG Learning · Not for redistribution

actual yield = 65 ÷ 100 × 12.0 = **7.8 g**

5.9 Atom Economy

Atom economy is a measure of the proportion of the reactants that end up as the **desired (useful) product**. A high atom economy means less waste.

atom economy = (relative formula mass of the desired product ÷ sum of relative formula masses of all reactants) × 100%

Use the balancing numbers from the equation. If there is only one product (an **addition** reaction), the atom economy is **100%**.

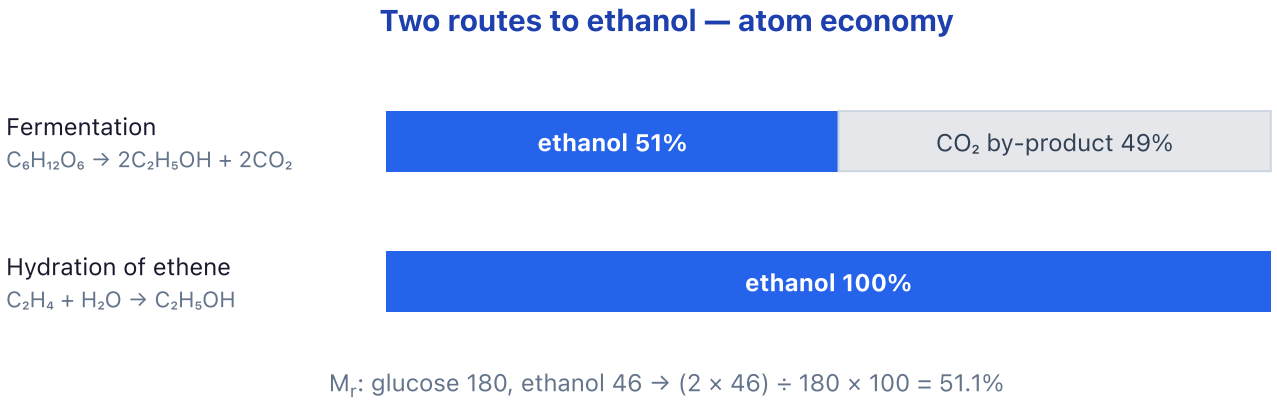
Worked Example: Calculate the atom economy for making calcium oxide by heating limestone: $CaCO_3 \rightarrow CaO + CO_2$ (M_r: $CaCO_3$ = 100, CaO = 56)

atom economy = 56 ÷ 100 × 100 = **56%** (the other 44% leaves as CO_2 waste)

Worked Example: Iron is produced by: $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$. Calculate the atom economy for iron. (A_r: Fe = 56, C = 12, O = 16)

M_r Fe_2O_3 = 160; 3 × M_r CO = 3 × 28 = 84 → total reactants = 244
desired product = 2 × 56 = 112
atom economy = 112 ÷ 244 × 100 = **45.9%**

© RAG Learning · Not for redistribution



M_r: glucose 180, ethanol 46 → (2 × 46) ÷ 180 × 100 = 51.1%

Choosing a reaction pathway (HT)

Industry chooses a route to a product by weighing up several factors, not just one:

Factor	What to look for
Atom economy	High — less waste, more efficient use of raw materials, more sustainable
Yield	High percentage yield — more product from the same reactants
Rate	Fast enough to make product in an acceptable time
Equilibrium position	For reversible reactions, conditions that give an acceptable proportion of product
Usefulness of by-products	If the by-product can be sold or used, a low atom economy matters less
Other	Cost and availability (renewable or finite) of raw materials, energy needed, safety

Worked Example: (HT) Compare the two routes to ethanol shown above.

Hydration of ethene: atom economy 100%, fast, continuous process and gives pure ethanol — but needs high temperature (~300 °C) and pressure (~60–70 atm) with a phosphoric acid catalyst, and ethene comes from crude oil, a **finite** resource.

© RAG Learning · Not for redistribution

Fermentation: atom economy only 51% and slow (batch process, takes days), producing a dilute solution that must be distilled — but uses **renewable** sugar from crops at low temperature (~30 °C), and the CO_2 by-product can be captured and sold.
Conclusion: where crude oil is cheap and plentiful, hydration is often chosen for its efficiency; where crops are plentiful and oil is expensive, fermentation may be preferred.

5.10 Molar Volume of Gases (HT)

One mole of **any gas** occupies the same volume at the same temperature and pressure. At **room temperature and pressure (RTP)** — 20 °C and 1 atmosphere — the **molar volume** of a gas is **24 dm³** (24 000 cm³).

volume of gas (dm³) = amount (mol) × 24 ↔ amount (mol) = volume (dm³) ÷ 24

volume of gas (cm³) = amount (mol) × 24 000

Worked Example: (HT) Calculate the volume of hydrogen produced at RTP when 0.48 g of magnesium reacts with excess hydrochloric acid. (A_r Mg = 24)

$Mg + 2HCl \rightarrow MgCl_2 + H_2$ (Mg : H_2 = 1 : 1)
moles Mg = 0.48 ÷ 24 = 0.020 mol → 0.020 mol H_2
volume H_2 = 0.020 × 24 = **0.48 dm³** (480 cm³)

Worked Example: (HT) What mass of calcium carbonate must react with excess acid to produce 600 cm³ of carbon dioxide at RTP? (M_r $CaCO_3$ = 100)

$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$ (1 : 1)
moles CO_2 = 600 ÷ 24 000 = 0.0250 mol → 0.0250 mol $CaCO_3$
mass = 0.0250 × 100 = **2.50 g**

© RAG Learning · Not for redistribution

Avogadro's law — reacting volumes of gases

Avogadro's law: equal volumes of all gases at the same temperature and pressure contain the **same number of molecules**. So for reactions between gases, the **ratio of volumes** is the same as the **mole ratio** in the balanced equation — you do not need to convert to moles at all.

Worked Example: (HT) Propane burns completely: $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$. What volume of oxygen is needed to burn 20 cm³ of propane, and what volume of carbon dioxide is produced? (All volumes at RTP.)

Volume ratio propane : oxygen : carbon dioxide = 1 : 5 : 3
oxygen = 5 × 20 = **100 cm³**
carbon dioxide = 3 × 20 = **60 cm³**
(The water is a liquid at RTP, so it has a negligible volume and is not included.)

Worked Example: (HT) 50 cm³ of carbon monoxide is burned in 40 cm³ of oxygen: $2CO(g) + O_2(g) \rightarrow 2CO_2(g)$. What is the total volume of gas at the end (RTP)?

50 cm³ CO needs 25 cm³ O_2 → oxygen is in excess by 40 − 25 = 15 cm³.
CO_2 formed = 50 cm³.
Total gas = 50 + 15 = **65 cm³**

Exam Tip: The 24 dm³ value is given on the exam paper when needed, but you must know when to use it: **only for gases**, and only at RTP. Watch the units — if the volume is in cm³ use 24 000.

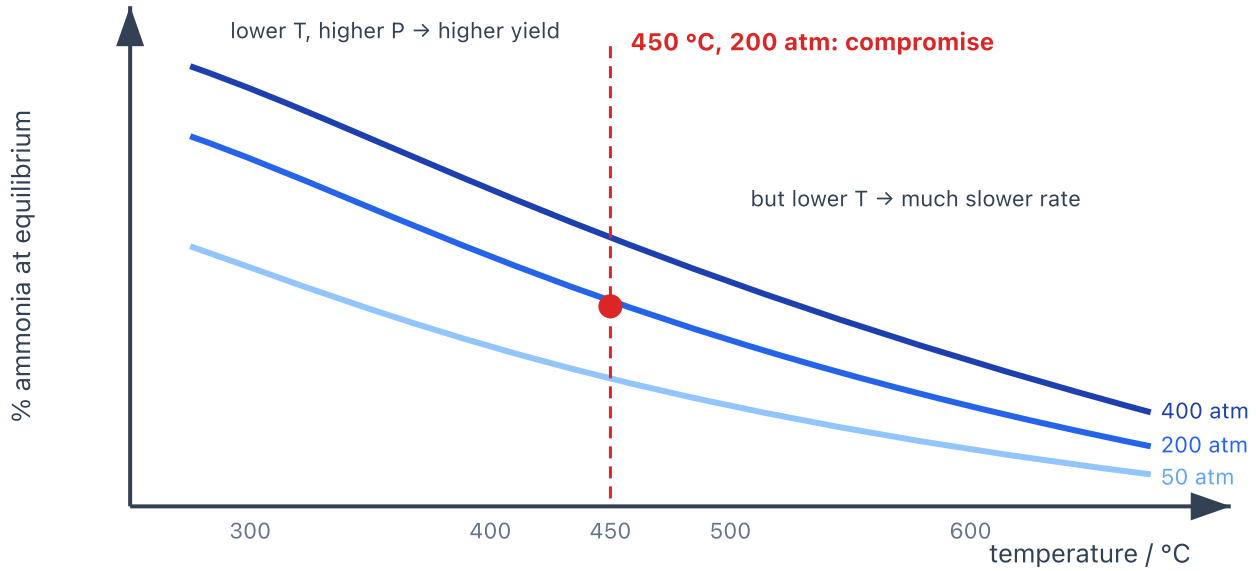
5.11 Dynamic Equilibria in Industry — the Haber Process

Recall from Topic 4 that the Haber process is a **reversible reaction** between nitrogen and hydrogen forming ammonia:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \quad \text{forward reaction exothermic}$$

(HT) Industrial chemists must choose conditions that give an **acceptable yield in an acceptable time** at an acceptable cost. The choice depends on the **availability and cost of raw materials and energy**, and on the **temperature, pressure and catalyst**.

Equilibrium yield of ammonia (shape of the trend)



Condition	Effect on yield	Effect on rate	Cost / practical issues	Choice
Temperature	Lower T gives a higher yield (forward reaction is exothermic)	Lower T makes the reaction too slow	Heating costs energy	**450 °C** — compromise between yield and rate
Pressure	Higher P gives a higher yield (4 gas molecules → 2)	Higher P increases rate (particles closer together)	Very high pressure needs expensive, thick-walled plant and pumps; more energy to compress gases; greater safety risk	**200 atm** — compromise between yield and cost/safety

© RAG Learning · Not for redistribution

Catalyst	No effect on yield	Increases rate — equilibrium reached faster	Cheap (iron); lasts a long time; allows a lower temperature	**Iron**

Rate of attainment of equilibrium **(HT)**

Changing the conditions affects how **quickly** equilibrium is reached, as well as (for temperature, pressure and concentration) where the equilibrium position lies:

- **Higher temperature** — particles have more energy and collide more often, so equilibrium is reached **faster** (even though, for the Haber process, the yield at equilibrium is lower).
- **Higher pressure** (gases) — particles are closer together, collisions are more frequent, so equilibrium is reached **faster**.
- **Higher concentration** of reactants — more frequent collisions, so equilibrium is reached **faster**.
- **Catalyst** — speeds up the forward and backward reactions equally, so equilibrium is reached **faster** but the position of equilibrium (yield) is unchanged.

Raw materials and energy **(HT)**

- **Nitrogen** comes from the air — abundant and free (but separating it uses energy).
- **Hydrogen** comes from natural gas (methane) — cheap but a **finite** fossil fuel.
- The reaction is exothermic: the heat released is used to **pre-heat** incoming gases, saving energy.
- Unreacted N_2 and H_2 are **recycled**, so over many passes almost all the raw material is converted, even though only about 15% reacts per pass.

> **Exam Tip:** **(HT)** For "explain why 450 °C is used", give both halves: "a **lower temperature** would give a **higher yield** because the forward reaction is exothermic, but the **rate** would be too slow; 450 °C gives an **acceptable yield in an acceptable time**." The same compromise idea applies to pressure (yield vs cost/safety).

© RAG Learning · Not for redistribution

5.12 Fertilisers

Fertilisers provide plants with essential elements to promote growth. The three main elements are **nitrogen (N)**, **phosphorus (P)** and **potassium (K)**. NPK fertilisers contain compounds of all three.

Fertilisers must be **soluble** in water so that plants can absorb the ions through their roots. Many nitrogen fertilisers are **ammonium salts** or **nitrates** made from ammonia.

Making ammonium nitrate

Ammonia (an alkali) reacts with **nitric acid** in a neutralisation reaction to form **ammonium nitrate**, which is particularly rich in nitrogen (it contains nitrogen in both ions):

ammonia + nitric acid → ammonium nitrate

$$NH_3(aq) + HNO_3(aq) \rightarrow NH_4NO_3(aq)$$

Worked Example: Calculate the percentage by mass of nitrogen in ammonium nitrate, NH_4NO_3. (A_r: N = 14, H = 1, O = 16)

$M_r = 14 + (4 \times 1) + 14 + (3 \times 16) = 80$

% N = $(2 \times 14) \div 80 \times 100$ = **35%**

Ammonium sulfate — laboratory vs industry

$$2NH_3(aq) + H_2SO_4(aq) \rightarrow (NH_4)_2SO_4(aq)$$

	Laboratory preparation	Industrial production
Raw materials	Bought ammonia solution and dilute sulfuric acid	Must be **made first** in several stages: ammonia by the Haber process (N_2

		from air, H_2 from natural gas); sulfuric acid by the Contact process (from sulfur, air and water)
Scale	**Small scale** — a few grams	**Very large scale** — thousands of tonnes
Method	**Titration**: titrate ammonia solution with acid using methyl orange to find the reacting volumes; repeat **without indicator**; evaporate some water, leave to **crystallise**, filter and dry	Ammonia gas reacted with concentrated sulfuric acid in large reaction vessels; the product is dried (spray-drying or crystallising)
Process type	**Batch**	**Continuous**
Energy	Heating to evaporate water	Reaction is exothermic — heat used to evaporate water; less energy input needed per tonne

Exam Tip: When comparing lab and industrial preparation, the specification's key words are **scale** (small vs much larger) and **stages** (in industry the raw materials ammonia and sulfuric acid have to be manufactured first). Use the words "batch" and "continuous" for extra credit.

5.13 Chemical Cells

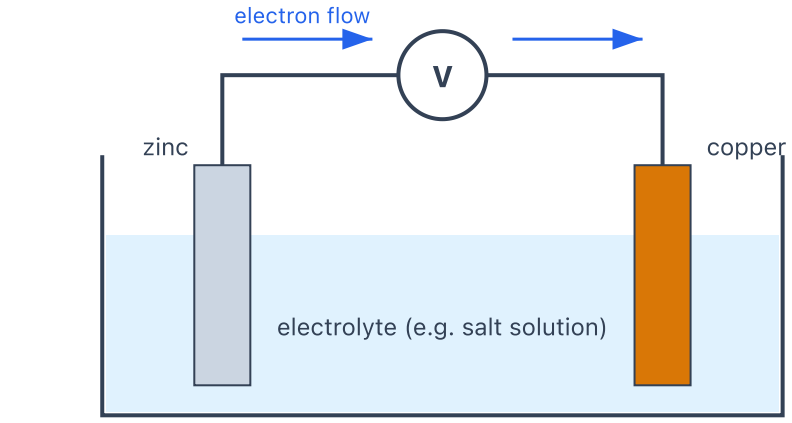
A **chemical cell** converts chemical energy into electrical energy. A simple cell is made by placing **two different metals** in an **electrolyte**. A chemical cell **produces a voltage until one of the reactants is used up**.

The more reactive metal loses electrons more readily, so electrons flow through the external circuit from the more reactive metal to the less reactive metal. The **bigger the difference in reactivity**

© RAG Learning · Not for redistribution

between the two metals, the **larger the voltage**. A **battery** is two or more cells connected in series.

A simple chemical cell



more reactive metal: loses electrons (−) less reactive metal (+)

Metals used	Approximate voltage
Magnesium / copper	Large (about 2 V)
Zinc / copper	Medium (about 1 V)
Iron / copper	Small (about 0.5 V)
Copper / copper	Zero — the metals must be different

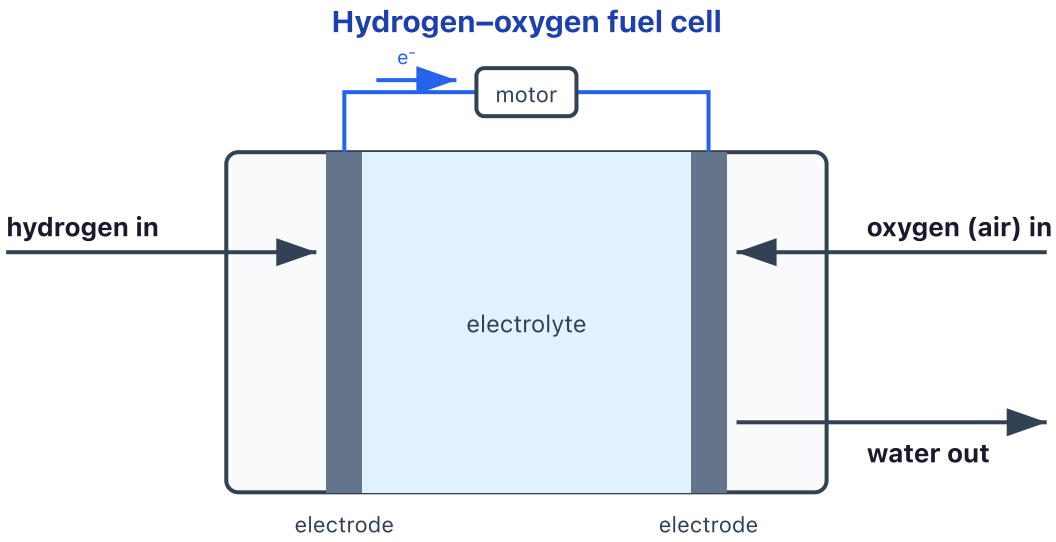
In non-rechargeable cells (e.g. alkaline batteries) the chemical reactions stop when one of the reactants has been used up and the cell is "flat". Rechargeable cells can be recharged because the reactions are reversed when an external current is supplied.

5.14 Hydrogen–Oxygen Fuel Cells

> In a **hydrogen–oxygen fuel cell**, hydrogen and oxygen are used to produce a **voltage**, and **water is the only product**. Unlike a chemical cell, the reactants (fuel and oxygen) are **supplied continuously**, so the fuel cell keeps producing electricity as long as they are supplied.

© RAG Learning · Not for redistribution

hydrogen + oxygen → water $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$



Evaluating fuel cells

Strengths	Weaknesses
The only product is **water** — no CO_2 or other pollutants at the point of use	**Hydrogen is a gas**: it takes up a large volume, so it must be stored under high pressure or as a liquid at very low temperature; tanks are heavy and bulky
More **efficient** than burning fuel in an engine — fewer energy-transfer stages	Hydrogen is **highly flammable/explosive**, so storage and transport carry risks
Produce a **continuous** supply of electricity as long as fuel is supplied — no long recharging time (quick refuelling)	Hydrogen is usually made from **natural gas** (releases CO_2) or by electrolysis of water, which needs electricity that may come from fossil fuels
Few moving parts, quiet, little maintenance	Fuel cells are **expensive** (use catalysts such as platinum); few hydrogen refuelling stations
Lighter than rechargeable batteries for the same energy stored in long-range	Fuel cells eventually wear out and contain materials that need careful disposal

© RAG Learning · Not for redistribution

use; no toxic metal waste when disposed of	

Worked Example: Evaluate the use of hydrogen fuel cells, rather than rechargeable lithium batteries, to power buses.

For fuel cells: buses can be refuelled in minutes rather than charged for hours, so they can run all day; the only exhaust is water, so there is no local air pollution; a large bus would need a very heavy battery pack.
Against: hydrogen must be stored in large high-pressure tanks and is flammable; hydrogen production often releases CO_2; refuelling infrastructure is limited and fuel cells are expensive.
Judgement: for buses running long routes from a single depot (where one hydrogen supply can be installed), fuel cells are a good choice; for short urban routes, batteries may be cheaper.

Exam Tip: "Water is the only product" is only true at the point of use. In evaluation questions, gain the higher marks by mentioning **where the hydrogen comes from** — if it is made using fossil fuels, CO_2 is still produced overall.

Key Points Summary

- **5.1C** — Most metals are transition metals: high melting point, high density, coloured compounds, catalytic activity (iron in the Haber process).
- **5.2C, 5.3C** — Corrosion is oxidation; rusting needs oxygen and water. Prevent it by excluding oxygen/water (paint, oil, grease, plastic, plating) or by sacrificial protection with a more reactive metal (zinc, magnesium; galvanising).
- **5.4C** — Electroplating: object = cathode, plating metal = anode, electrolyte contains ions of the plating metal; improves appearance and corrosion resistance.
- **5.5C, 5.6C** — Alloys are stronger because different-sized atoms distort the layers so they cannot slide; iron is alloyed (e.g. with chromium/nickel → stainless steel) to make alloy steels.

© RAG Learning · Not for redistribution

- **5.7C** — Relate uses to properties: aluminium, copper, gold, magnalium (Al + Mg), brass (Cu + Zn).
- **5.8C (HT)** — concentration (mol dm^{-3}) = mol ÷ dm^3; convert g dm^{-3} ↔ mol dm^{-3} using M_r.
- **5.9C (CP5)** — Accurate titration: pipette, burette, indicator, white tile, dropwise near end-point, concordant titres within 0.10 cm^3.
- **5.10C (HT)** — Calculate an unknown concentration or volume from titration results, using the mole ratio.
- **5.11C, 5.12C** — Percentage yield = actual ÷ theoretical × 100; losses from incomplete reactions, practical losses, side reactions.
- **5.13C, 5.14C** — Atom economy = M_r desired product ÷ total M_r reactants × 100%. **5.15C (HT)** — choose a pathway using atom economy, yield, rate, equilibrium position and by-products.
- **5.16C–5.18C (HT)** — Molar gas volume 24 dm^3 at RTP; mass ↔ gas volume calculations; Avogadro's law: gas volume ratio = mole ratio.
- **5.19C** — The Haber process is a reversible reaction between nitrogen and hydrogen forming ammonia.
- **5.20C, 5.21C (HT)** — Higher temperature, pressure or concentration, or a catalyst, make equilibrium reached faster; Haber conditions (450 °C, 200 atm, iron) are a compromise of yield, rate, cost of raw materials and energy, and safety.
- **5.22C–5.24C** — NPK fertilisers; $NH_3 + HNO_3 \rightarrow NH_4NO_3$; compare lab (titration, small, batch) and industrial (multi-stage, large, continuous) production of ammonium sulfate.
- **5.25C–5.27C** — A chemical cell gives a voltage until a reactant is used up; a hydrogen–oxygen fuel cell produces a voltage with water as the only product; evaluate the strengths and weaknesses of fuel cells.

© RAG Learning · Not for redistribution